

KnowSelf

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

arxiv.org/abs/2504.03553

For ML Researchers & Graduate Students

The Problem: 'Flood Irrigation' in Agent Training

Current Paradigms

1. Trajectory Imitation (ReAct)

Blindly follows gold trajectories without situational judgment

2. Trial-and-Error (Reflexion)

Relies on repeated failures to self-correct, inefficient

3. Knowledge-Augmented (KnowAgent, WKM)

Injects domain knowledge at every step — wasteful

Human Self-Awareness

Can I handle this directly?

→ Yes: Act now (Fast Thinking)

→ No: Should I reflect?

→ Yes: Think it through (Slow Thinking)

→ No: Seek knowledge (Knowledgeable Thinking)

KnowSelf brings situational self-awareness to LLM agents — enabling autonomous knowledge regulation

Three Situational Thinking Modes



Agent Decision Step



Decision Criteria:



Fast:

Predicted action = Gold action



Slow:

Reflection fixes incorrect action



Knowledgeable:

Even reflection fails → consult knowledge

Special Tokens:

[FAST]

→ Execute action directly

[SLOW]

→ Trigger self-reflection

[KNOW]

→ Retrieve from knowledge base

KnowSelf Framework: Data → Training → Inference

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1. Data Construction

Agent self-explores environment
Heuristic labels each step with:

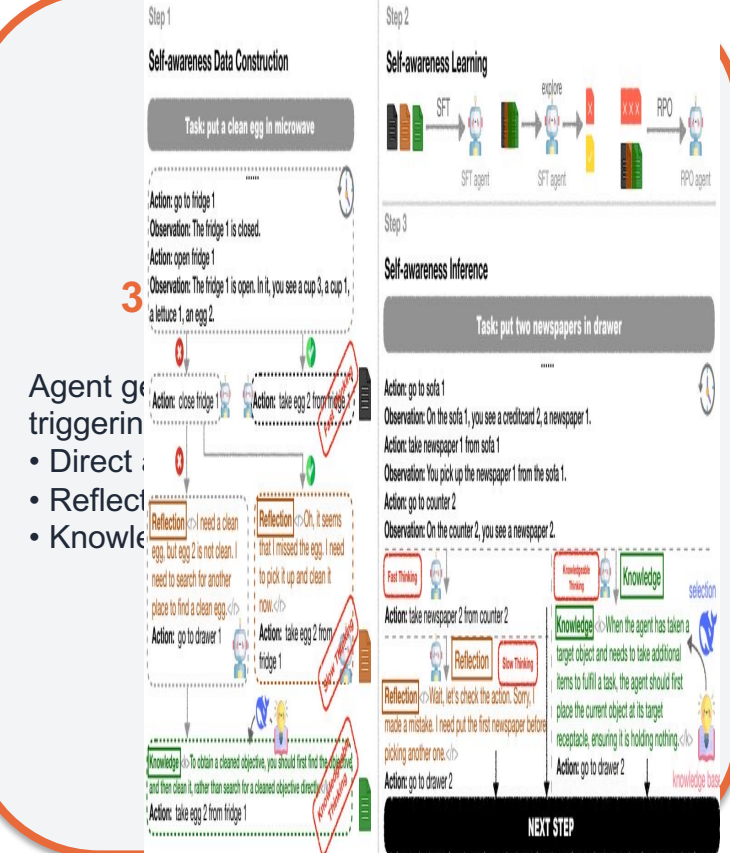
- [FAST] if action matches gold
- [SLOW] if reflection fixes it
- [KNOW] if external knowledge needed

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2. Two-Stage Training

Stage 1: Supervised Fine-Tuning
→ Learn initial self-awareness

Stage 2: RPO (Reward-based Policy Optimization)
→ Refine token generation



Qualitative Examples: KnowSelf Outperforms Frontier Models

Task: Put a hot mug in cabinet

OpenAI-O1: Incorrectly tries to take an egg from the microwave after opening it — fails to realize mug needs heating.

KnowSelf (Llama-8B):

- Uses [SLOW] token to trigger reflection:
- Recognizes mug is not hot
 - Realizes must heat mug first using microwave
 - Successfully completes task

Task: Put a saltshaker in drawer

DeepSeek-R1: Reflects that drawer is open and tries to close it first — which fails.

KnowSelf (Llama-8B):

- Uses [KNOW] token to retrieve knowledge:
- Agent should place object directly
 - No need to remove unrelated items
 - Completes task correctly

(a) Task: put a hot mug in cabinet

OpenAI-O1

Action: take mug 1 from countertop 1
Observation: You pick up the mug 1 from the countertop 1.
Action: go to microwave 1
Observation: The microwave 1 is closed.
Action: open microwave 1
Observation: You open the microwave 1. In it, you see an egg 1.
[Knowledge]<=>If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of placing the object on it.</=>
Action: take egg 1 from microwave 1
Observation: Nothing happens.

KnowSelf (Llama-8B)

Action: take mug 1 from countertop 1
Observation: You pick up the mug 1 from the countertop 1.
Action: go to microwave 1
Observation: The microwave 1 is closed.
Action: open microwave 1
Observation: You open the microwave 1. In it, you see an egg 1.
Action: put mug 1 in/on microwave 1
[Reflection]<=>Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.</=>
Action: heat mug 1 with microwave 1

(b) Task: put a saltshaker in drawer

DeepSeek-R1

Action: take saltshaker 1 from countertop 3
Observation: You pick up the saltshaker 1 from the countertop 3.
Action: go to drawer 1
Observation: The drawer 1 is open. In it, you see a fork 1.
Action: put saltshaker 1 in/on drawer 1
[Reflection]<=>Wait, the drawer is open, so the saltshaker might fall inside. I need to close the drawer first.</=>
Action: close drawer 1
Observation: You close the drawer 1.
Action: put saltshaker 1 in/on drawer 1
Observation: Nothing happens.

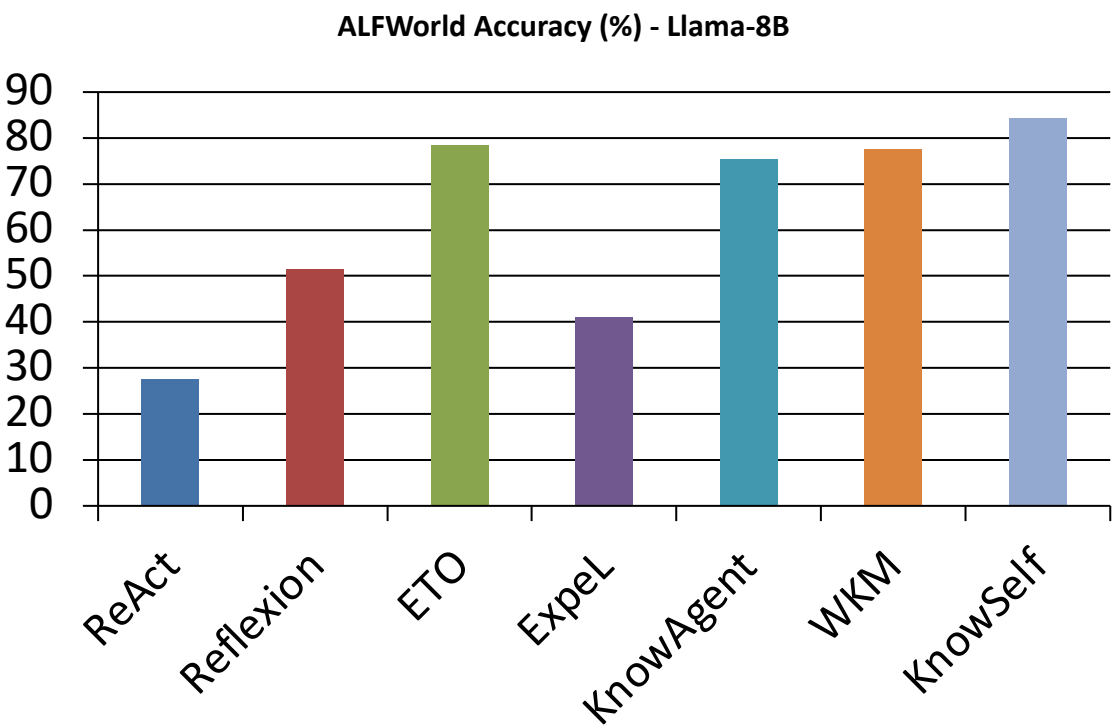
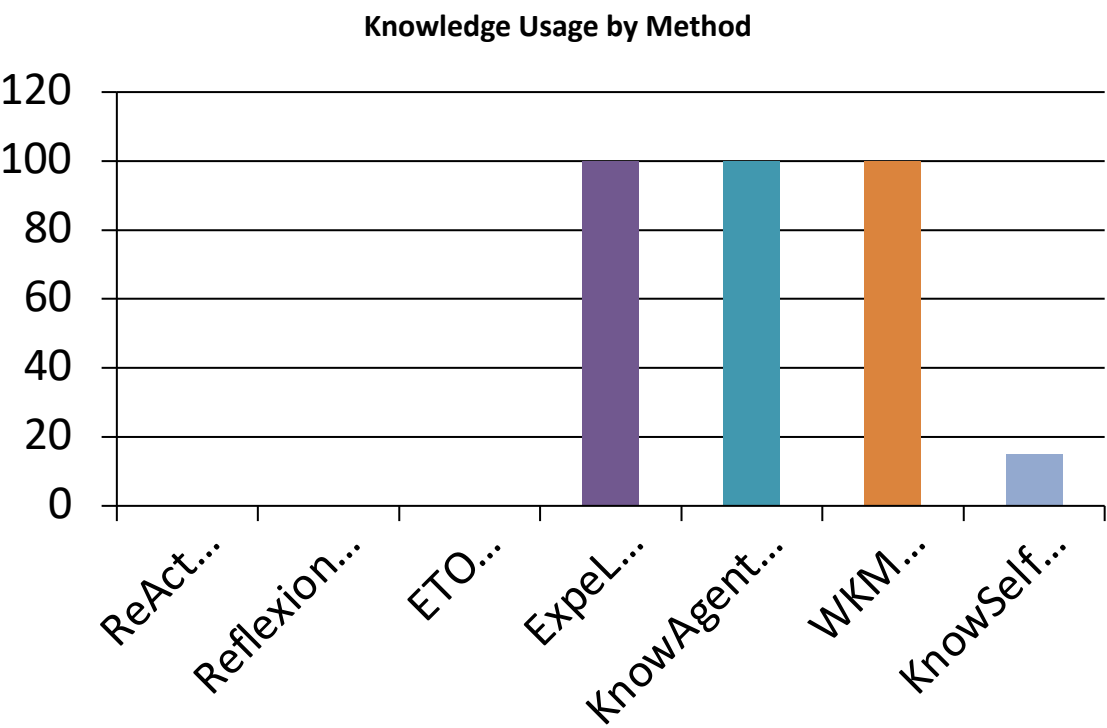
KnowSelf (Llama-8B)

Action: take saltshaker 1 from countertop 3
Observation: You pick up the saltshaker 1 from the countertop 3.
Action: go to drawer 1
Observation: The drawer 1 is closed.
Action: open drawer 1
Observation: You open the drawer 1. In it, you see a fork 1.
[Knowledge]<=>When the agent encounters a situation where receptacles contain other items but only needs to place the target object, the agent should proceed to place the object without removing unrelated items.</=>
Action: put saltshaker 1 in/on drawer 1

ALFWorld Results: KnowSelf Achieves 84.33% with Only 15% Knowledge Usage

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Minimal Knowledge, Maximum Impact: Efficiency Advantage



KnowSelf uses 15x less knowledge than 100%-knowledge methods, yet achieves highest accuracy (84.33%)

Even Gemma-2B KnowSelf (26% knowledge) reaches 79.85% — matching GPT-4o Expel and exceeding all other Gemma-2B baselines

Key Takeaways: Self-Awareness Is the Missing Piece

1

Indiscriminate knowledge injection is wasteful and suboptimal

Flood irrigation approaches (100% knowledge) often underperform selective knowledge use

2

Agents can learn situational self-awareness via special tokens

Lightweight heuristic labeling on self-explored trajectories — no expensive human annotation

3

KnowSelf with Llama-8B (84.33%) surpasses GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%)

Demonstrates that selective knowledge use combined with self-awareness outperforms blanket application

4

Only 15–26% knowledge usage needed versus 100% for competing methods

5–6x reduction in knowledge queries while achieving best-in-class results

5

The paradigm generalizes across model scales

~~Gemma 2B KnowSelf (79.85%) matches GPT-4o ExpeL, exceeding all Gemma 2B baselines~~

Future Directions: Extend to multi-agent scenarios, apply to reasoning tasks (MATH, GPQA), investigate dynamic knowledge base integration